

Query 5: Which customer profile churns the most?

```
SELECT
  c5 AS Contract,

  CASE
    WHEN c6 <= 12 THEN '0–12 months'
    WHEN c6 <= 24 THEN '13–24 months'
    WHEN c6 <= 48 THEN '25–48 months'
    ELSE '48+ months'
  END AS tenure_group,

  CASE
    WHEN c19 < 35 THEN 'Low'
    WHEN c19 < 70 THEN 'Medium'
    ELSE 'High'
  END AS charge_group,

  COUNT(*) AS total_customers,

  SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,

  ROUND(
    100.0 * SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) / COUNT(*),
    2
  ) AS churn_rate_percent

FROM WA_FnUseC_TelcoCustomerChurn
WHERE c21 != 'Churn'
GROUP BY c5, tenure_group, charge_group
ORDER BY churn_rate_percent DESC;
```

Query 4: Does monthly charge amount predict

```
SELECT
  CASE
    WHEN c19 < 35 THEN 'Low (<35)'
    WHEN c19 < 70 THEN 'Medium (35–70)'
    ELSE 'High (70+)'
  END AS charge_group,

  COUNT(*) AS total_customers,

  SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,

  ROUND(100.0 * SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
churn_rate_percent

FROM WA_FnUseC_TelcoCustomerChurn
```

```
WHERE c21 != 'Churn'
GROUP BY charge_group;
```

Query 3: How does tenure affect churn?

```
SELECT
  CASE
    WHEN c6 <= 12 THEN '0-12 months'
    WHEN c6 <= 24 THEN '13-24 months'
    WHEN c6 <= 48 THEN '25-48 months'
    ELSE '48+ months'
  END AS tenure_group,

  COUNT(*) AS total_customers,

  SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,

  ROUND(100.0 * SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
churn_rate_percent

FROM WA_FnUseC_TelcoCustomerChurn
WHERE c21 != 'Churn'
GROUP BY tenure_group;
```

Query 2: Churn rate by contract type:

```
SELECT
  c5 AS Contract,
  COUNT(*) AS total_customers,
  SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,
  ROUND(100.0 * SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
churn_rate_percent
FROM WA_FnUseC_TelcoCustomerChurn
WHERE c21 != 'Churn'
GROUP BY c5;
```

Query 1: What is the overall churn rate?

```
SELECT
  COUNT(*) AS total_customers,
  SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) AS churned_customers,
  ROUND(100.0 * SUM(CASE WHEN c21 = 'Yes' THEN 1 ELSE 0 END) / COUNT(*), 2) AS
churn_rate_percent
FROM WA_FnUseC_TelcoCustomerChurn
WHERE c21 != 'Churn';
```

FINDINGS:

- Customers on month-to-month contracts have the highest churn rate, significantly exceeding one-year and two-year contracts, indicating that contract length is a major driver of retention.
- Tenure has a strong inverse relationship with churn: customers in their first 12 months are the most likely to leave, while long-term customers (48+ months) have the lowest churn rates.
- Higher monthly charges are associated with increased churn, suggesting that price sensitivity plays a role in customer attrition.
- Overall, the highest-risk customer profile consists of new customers on month-to-month contracts with higher monthly charges, making this group the primary driver of churn in the dataset.